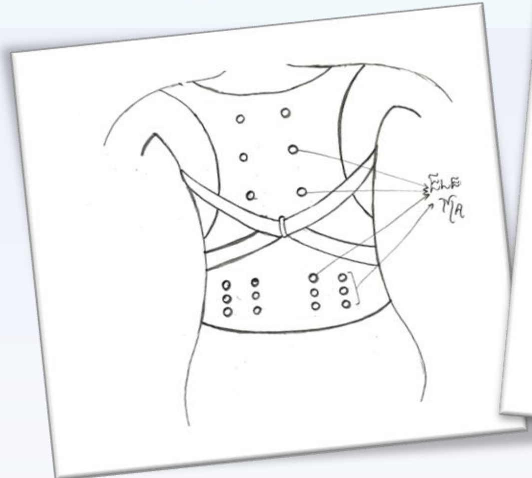
MEDHA 2024

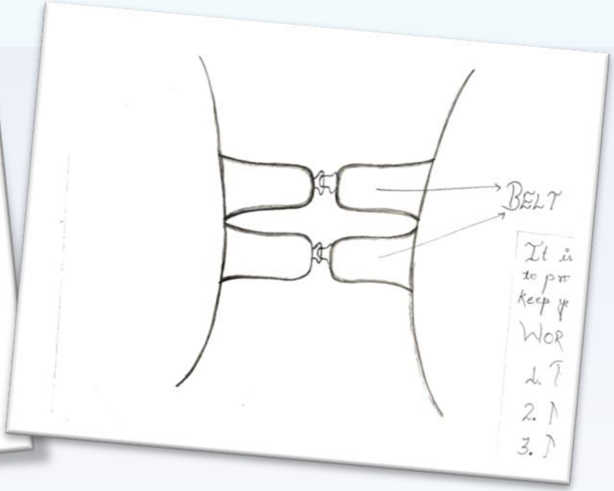
Medical Device Hackathon

23rd - 24th September, 2024

PosturePro Belt

"Belt for Physiotherapist"





Project Author

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Age	**20**
Location	**GHAZIABAD**
Mother Tongue	**HINDI**

Project Title - “PosturePro Belt”

Problem Background

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Domain of Proposed Problem	• **Physiotherapy and Ergonomics:** Prevents and alleviates back pain in caregivers during patient-assisted physical exercises. • **Wearable Medical Device:** Portable, smart belt with posture correction and pain relief functionality. • **Sensor-Drive along TENS:** Uses sensors and electromagnets to detect and relieve pain in real-time. • **Comfort and Flexibility:** Designed for comfort with flexible foam and secure elastic straps, ensuring ease of movement.
Physiology of Anatomy in Normal Condition	• **Structural Organization**: The body consists of systems (skeletal, muscular, nervous) that work together. • **Homeostasis**: Physiological processes maintain stable internal conditions (temperature, pH). • **Tissue Functionality**: Different tissues (epithelial, connective, muscle, nervous) perform specialized roles. • **System Interactions**: Organ systems collaborate for efficient physiological processes (respiration, circulation). • **Adaptation and Response**: The body adapts to physical activity and environmental changes. • **Nutrient and Waste Management**: Efficient processing of nutrients and waste is vital for cellular function.
Physiology of Anatomy in Abnormal/diseased Condition	• **Structural Changes**: Diseases can lead to alterations in anatomical structures, affecting the function of organs and systems. • **Homeostatic Imbalance**: Disruptions in physiological processes result in the inability to maintain stable internal conditions, leading to conditions like fever or dehydration. • **Tissue Dysfunction**: Abnormalities in tissue types can impair their specific functions, contributing to symptoms and disease progression. • **Inadequate Response to Stress**: The body may struggle to adapt to physical stressors, resulting in increased susceptibility to injury or illness.

Analysis

#	Key activities by Author	Difficulties faced by Author	Value Additions
1	Sensor (Hardware) Integration, PPT making.	Sensors are not available as per the requirement, some of the slides in ppt require information which we are not able to give at this stage.	Got to work on a diverse environment with dynamic team.
2	Prototyping, clinical assessment, coordinating design innovation.	Design limitations, clinical validation challenges.	Enhanced patient care.
3	Component sourcing, supplier coordination, collaboration on design	Sourcing components, managing product availability.	Timely prototyping and development.
4	Overseeing the mechanical design.	Material constraints.	Support for functional optimization in the device.
5	Collaborating with other team members.	Integrating electronic components with mechanical designs.	Improved device performance.
6	Creating design layouts on chart paper.	Handling task delegation, and maintaining project organization	Visualizing designs through layout creation.

Existing & Emerging Solutions

Device Name	Brand Name	Website URL	Key Features	Limitations/gaps
Lumbo Sacral Belt	VISSCO neXt	https://vissconext.com/products/lumbocare-lumbo-sacral-belt	This lower back support belt can be used for acute lower back pain, slip disc, disc herniation, lumbar spondylosis, muscle weakness in the lumbar spine and other conditions causing pain and discomfort in the lower back, and requiring mild support.	This is not especially designed for physiotherapist.
Lace Pull Lower Back Support	Hansaplast	https://www.hansaplastindia.com/products/heat-plaster/lace-pull-lower-back-support	lower back support with lace pull mechanism designed to provide optimal support along with wearing comfort to give long-lasting pain relief so you can continue to stay active and live life uninterrupted.	It does not relief neck and shoulder pain.

Relevant Papers/Publications

#	Title	Authors	Year of Publication	Journal Name	Key Takeaways	Keywords
1	Effectiveness of a Lumbar Belt in Subacute Low Back Pain	Calmels, Paul, Queneau, Patrice Hamonet, Claude	2008	Spine	Multicentric, randomized, and controlled study of clinical evaluation of medical device in subacute low back pain. Objective. To evaluate the effects of an elastic lumbar belt on functional capacity, pain intensity in low back pain treatment, and the benefice on medical cost.	Lumbar belt wearing is consequent in subacute low back pain to improve significantly the functional status, the pain level, and the pharmacologic consumption
2	Influence of elastic lumbar support belts on trunk muscle function in patients with non-specific acute lumbar back pain	Christoph Anders, Agnes Hübner.	2019	PubMed Central	Treatment for acute nonspecific back pain, elastic back support belts, are valued for their ability to accelerate natural self-healing, but there are concerns of a deconditioning effect due to their reliance on passive stabilization	For the study, only 36 people could be included in the analysis, which makes broad generalization from the results impossible.

Prior Art

Relevant Patents

#	Title	Year of Filing	Countries protected	Key Takeaways/Claims	Patent Filing Status
1	Belt for relieving back pain	2014	US	Back straightening devices with shoulder braces to force back the shoulder to obtain a correct curvature of the spine	Abandoned
2	Support for the backbone or spinal column of the human body	1939	US	Orthopaedic devices, e.g. splints, casts or braces	Expired - Lifetime

Applicable Regulatory standards

#	Standard No.	Title of the Standard	Purpose
1	ISO 13485	Medical Devices - Quality Management Systems	Ensures consistent design, development, and production quality of medical devices.
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4	ISO 13485	Medical Devices - Quality Management Systems	Ensures consistent design, development, and production quality of medical devices.
	Available Testing Labs		
1	**BSI Group** Focuses on medical device testing and certification, particularly for CE marking and ISO 13485.		

Need & Market Analysis

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#	Market Size
	➢ **Growing Market**: Rapid increase in demand due to heightened awareness of spinal health. ➢ **Projected Growth**: Global back pain treatment market expected to reach billions. ➢ **High Demand**: Need for wearable devices like belts and braces, especially for caregivers. ➢ **Opportunity**: Significant potential for innovative, multi-functional products.

#	Gaps in the Existing Solutions
	❑ **Limited Focus**: Current products prioritize posture alignment but lack real-time pain relief features. ❑ **Rigid Design**: Many devices are uncomfortable and not user-friendly. ❑ **Need for Integration**: Demand for solutions that combine posture correction with pain management. ❑ **Market Gap**: Clear opportunity for comprehensive, multi-functional devices.

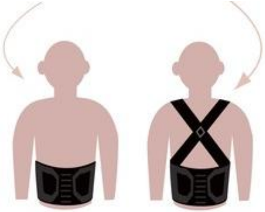
Problem Definition and Intended Use

Need Statement	**There is a need for a device that provides effective spine and posture correction while also offering pain relief for caregivers and individuals experiencing back strain.**
Who (User of the Device)	Caregivers, healthcare professionals, and individuals with back pain
What	A prototype belt for spine and posture correction with integrated pain relief features.
When	During physical therapy sessions, daily activities, or prolonged sitting/standing.
Where	Homes, clinics, and rehabilitation centers.
Why	To enhance comfort, prevent back pain, and improve overall spinal health and posture.
Intended Use	To correct posture, provide pain relief, and support individuals during physical therapy and daily activities.

User & Functional Requirements

#	User Requirements	Functional Requirements	Reference
1	The device should provide real-time feedback on posture and spine alignment.	The belt must include sensors to monitor and analyze posture in real-time.	ISO 13485: Medical Devices - Quality Management Systems
2	The device should offer adjustable settings for different body sizes.	The belt should have adjustable straps for a customized fit.	IEC 60601-1: Medical Electrical Equipment - Safety
3	The device should provide pain relief through massage or other therapeutic methods.	The device must include an integrated massage feature triggered by sensors.	ISO 14971: Medical Devices - Application of Risk Management
4	The device should be comfortable for extended wear.	The belt must be made from flexible, breathable materials for user comfort.	ISO 10993: Biological Evaluation of Medical Devices

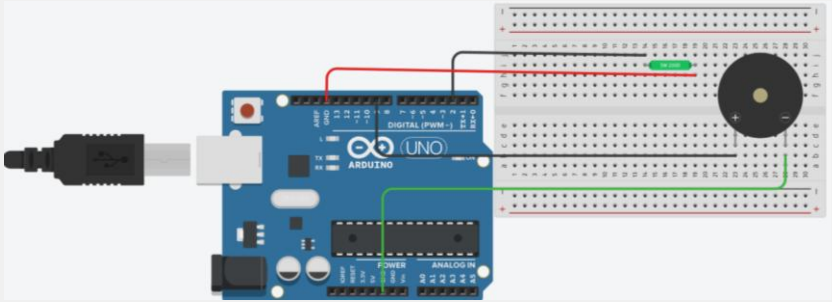


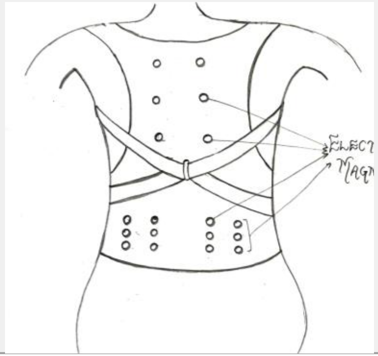



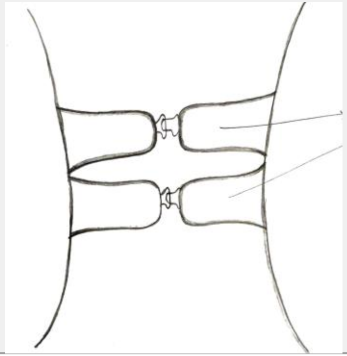
Potential Concepts

Potential Concepts to Solve the Problem

Real Time Monitoring	Innovative Design







#	Key Features of Proposed Device
	❑ **Posture Monitoring**: Equipped with advanced sensors that continuously track spinal alignment and posture in real-time. ❑ **Feedback Mechanism**: Provides immediate alerts and feedback to users through a buzzer, promoting awareness and corrective actions.

#	Value Propositions for Proposed Device
	❑ **Improved Posture**: Promotes proper spinal alignment, reducing the risk of long-term musculoskeletal issues. ❑ **Enhanced Comfort**: Provides significant relief from back pain and discomfort, improving daily life for caregivers.

Tentative Development Cost

#	Electronics & Software			
	Arduino Uno, Buzzer, Jumper wire, MPU 6050, Breadboard, TENS, Wires	₹658		
#	**Mechanical**			
	For the clothing material.	Around ₹300		
#	**Testing**	YES		
	Prototype Testing	Done		
#	**Human Resource**			
#	**Contingency**			